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II PUC PHYSICS

IMPORTANT QUESTIONS

TARGET 50 MARKS

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EXAM STRATEGY — What to Cover

Study only the marks listed below for each chapter. Everything else can be safely skipped.

Chapter	Marks to Cover
Ch 1 – Electric Charges and Fields	2M ♦ 3M ♦ 5M
Ch 2 – Electrostatic Potential & Capacitance	2M ♦ 3M ♦ 5M
Ch 3 – Current Electricity	2M ♦ 5M ♦ Numericals
Ch 4 – Moving Charges and Magnetism	2M ♦ 3M ♦ 5M ♦ Nums
Ch 5 – Magnetism and Matter	2M ♦ 3M
Ch 6 – Electromagnetic Induction	2M ♦ 3M
Ch 7 – Alternating Current	Numericals only
Ch 8 – Electromagnetic Waves	2M only
Ch 9 – Ray Optics and Optical Instruments	3M only
Ch 10 – Wave Optics	✗ SKIP
Ch 11 – Dual Nature of Radiation and Matter	3M only
Ch 12 – Atoms	2M only
Ch 13 – Nuclei	3M only
Ch 14 – Semiconductor Electronics	2M ♦ 5M

PRIORITY ORDER — Complete in This Sequence

Start from Priority #1 and work downward. Don't skip to the next chapter until the current one is done.

Priori ty	Chapter	Focus
#1	Electric Charges and Fields	2M + 3M + 5M
#2	Electrostatic Potential and Capacitance	2M + 3M + 5M
#3	Current Electricity	2M + 5M + Numericals
#4	Moving Charges and Magnetism	2M + 3M + 5M + Nums
#5	Alternating Current	Numericals only
#6	Semiconductor Electronics	2M + 5M
#7	Electromagnetic Induction	2M + 3M
#8	Magnetism and Matter / EM Waves	2M + 3M / 2M only

⚠ **Note:** Chapter 10 (Wave Optics) is completely skipped. Ch 9 & Ch 11 — 3 marks only. Ch 12 — 2 marks only. Ch 7 — Numericals only.

II PUC PHYSICS — IMPORTANT QUESTIONS (TARGET 50)

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5 Marks Questions — Part 1

Chapter 1 — Electric Charges and Fields

1. Derive an expression for the electric field at a point on the axis of an electric dipole. (Pg 11)
2. Derive an expression for the electric field at a point on the equatorial line of an electric dipole. (Pg 12)
3. Using Gauss's law, arrive at the expression for electric field due to an infinitely long straight uniformly charged wire. (Pg 15)
4. Using Gauss's law, arrive at the expression for electric field at a point outside the charged spherical conducting shell. Also write the expression for electric field at a point on the surface of the same charged spherical shell. (Pg 17)

► **Problems**

► *Numerical Problems – refer textbook.*

Chapter 2 — Electrostatic Potential and Capacitance

5. Derive an expression for electric potential at a point due to a point charge. (Pg 32)

► **Problems**

► *Numerical Problems – refer textbook.*

Chapter 3 — Current Electricity

6. Using Kirchhoff's laws, arrive at the balancing condition of Wheatstone network. (Pg 78)
7. Obtain the expressions for equivalent emf and equivalent internal resistance of two cells of different emfs and different internal resistances connected in series. (Pg 75)
8. Obtain the expressions for equivalent emf and equivalent internal resistance of two cells of different emfs and different internal resistances connected in parallel. (Pg 76)
9. Assuming the expression for drift velocity, derive the expression for conductivity of a material. (Pg 69)

► **Problems**

► *Numerical Problems – refer textbook.*

Chapter 4 — Moving Charges and Magnetism

10. Derive an expression for the magnetic field at a point on the axis of a circular current loop using Biot-Savart's law. (Pg 94)
11. Derive the expression for force per unit length on two infinitely long straight parallel conductors carrying currents and hence define 'Ampere'. (Pg 96)
12. (i) State and explain Ampere's circuital law. (Pg 95) (ii) Derive the expression for the magnetic field at a point due to an infinitely long straight current carrying wire using Ampere's circuital law. (Pg 96)

Chapter 7 — Alternating Current

► **Problems only**

► *Numerical Problems – refer textbook.*

5 Marks Questions — Part 2

Chapter 9 — Ray Optics and Optical Instruments

13. Derive the expression for refractive index of the material of a prism in terms of angle of the prism and angle of minimum deviation. (Pg 16)
14. Derive lens maker's formula. (Pg 13)

► **Problems**

► *Numerical Problems – refer textbook.*

Chapter 10 — Wave Optics

15. (i) Two coherent waves of same amplitude and constant phase difference undergo interference. Obtain an expression for resultant displacement. (Pg 41) (ii) Mention the two uses of polaroids. (Pg 51)
16. (i) What is a wavefront? (Pg 36) (ii) Derive Snell's law for a plane wave undergoing refraction using Huygens principle. (Pg 37)
17. State Huygens' principle. Using Huygens principle show that the angle of incidence is equal to angle of reflection. (Pg 36)

► **Problems**

► *Numerical Problems – refer textbook.*

✗ **Note:** Chapter 10 is in the original question bank but is NOT in your exam scope. Skip during exam preparation.

Chapter 14 — Semiconductor Electronics

18. What is meant by rectification? Explain the working of a p-n junction diode as a full-wave rectifier, with the help of relevant circuit diagram and input-output waveforms. (Pg 108)
 19. (i) Explain the formation of energy bands in solids. (ii) On the basis of energy bands, distinguish between a conductor, a semiconductor and an insulator. (Pg 95)
 20. What is a half-wave rectifier? Draw a neat labelled circuit diagram and input-output waveforms for half-wave rectifier. Explain the working of half-wave rectifier. (Pg 107)
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3 Marks Questions

Chapter 1 — Electric Charges and Fields

21. Define linear charge density. Mention the expression for electric field at a point due to an infinite line of charge and explain the terms. (Pg 14 and 15)
22. Derive an expression for torque acting on an electric dipole placed in a uniform electric field. (Pg 13)
23. Give Coulomb's law in vector form and explain the terms. Define SI unit of charge. (Pg 5)
24. Write three properties of electric charge. (Pg 3)
25. State and explain Coulomb's law. (Pg 5)
26. Write the properties of electric field lines. (Pg 9)

Chapter 2 — Electrostatic Potential and Capacitance

27. Obtain the expression for equivalent capacitance of two capacitors connected in series. (Pg 47)
28. Obtain the expression for equivalent capacitance of two capacitors connected in parallel. (Pg 46)
29. What is a capacitor? Mention any two factors on which capacitance of a parallel plate capacitor depends. (Pg 47 and Pg 46)
30. Mention any three important results regarding electrostatics of conductors.

Ans: Inside a conductor electric field is zero.

At the surface of a charged conductor, electrostatic field is always normal to the surface at every point.

The interior of a conductor can have no excess charge in the static situation.

Electrostatic potential is constant throughout the volume of the conductor and has the same value on the surface.

Electric field at the surface of a charged conductor $E = \sigma/\epsilon_0 \hat{n}$

Electrostatic shielding.

31. Derive the relation between electric field and electric potential. (Pg 36)
32. Derive an expression for capacitance of a parallel plate capacitor separated by air or vacuum. (Pg 44)

Chapter 4 — Moving Charges and Magnetism

33. Explain the conversion of a galvanometer into an ammeter using relevant circuit diagram and expression. (Pg 102)
34. Explain the conversion of a galvanometer into a voltmeter using relevant circuit diagram and expression. (Pg 103)
35. In a chamber, a uniform magnetic field of 6.5×10^{-4} T is maintained. An electron is shot into the field with a speed of 4.8×10^6 m s⁻¹ normal to the field. Determine the radius of the circular orbit. Given: $e = 1.6 \times 10^{-19}$ C and $m_e = 9.1 \times 10^{-31}$ kg.
36. What is Lorentz force? Write the expression representing this force and explain the terms. (Pg 90)
37. What is a solenoid? Mention an expression for magnetic field at a point inside a solenoid and explain the terms. (Pg 98)

Chapter 5 — Magnetism and Matter

38. Write any three differences between diamagnetic and paramagnetic materials. (Pg 112)
39. State and explain Gauss's law in magnetism. Give its significance. (Pg 109)
40. Mention any three properties of magnetic field lines. (Pg 109)

41. Mention the properties of ferromagnetic materials. (Pg 113)

Chapter 6 — Electromagnetic Induction

42. Derive the expression for the motional emf induced in a conducting rod moving perpendicular to a uniform magnetic field. (Pg 119)
43. State and explain Lenz's law. (Pg 118)
44. Derive the expression for the magnetic energy stored in a coil, $U = \frac{1}{2}LI^2$. (Pg 125)
45. Describe the coil and bar magnet experiment to demonstrate the phenomenon of electromagnetic induction. (Pg 115)

Chapter 9 — Ray Optics and Optical Instruments

46. Show that the focal length of a spherical mirror is equal to half of its radius of curvature. (Pg 4)
47. Write the conditions of total internal reflection. Write any one application of total internal reflection. (Pg 10)
48. Write the uses of optical fibres. On what principle does an optical fibre work? (Pg 11)
49. Derive the expression for effective focal length of two thin lenses in contact. (Pg 15)
50. Draw the ray diagram for the formation of image by a compound microscope. Write the expression for linear magnification produced by the objective of compound microscope in terms of its tube length. (Pg 19)
51. Give the Cartesian sign convention used to measure distances and heights in spherical mirrors. (Pg 3)

Chapter 11 — Dual Nature of Radiation and Matter

52. Mention experimental observations of photoelectric effect. (Pg 61)
53. The work function of caesium is 2.14 eV. Calculate the threshold frequency for caesium.
54. What are matter waves? Explain the de Broglie hypothesis of matter waves. (Pg 64)
55. Write any three properties of a photon. (Pg 63)
56. Give Einstein's explanation of photoelectric effect.

Chapter 13 — Nuclei

57. Mention three features of nuclear force. (Pg 89)
58. Differentiate between nuclear fission and nuclear fusion. (Pg 90)
59. Calculate the binding energy of an alpha (α) particle in MeV from the following data. Mass of α -particle = 4.00260 u, Mass of neutron = 1.008662 u, Mass of proton = 1.007825 u.

2 Marks Questions

Chapter 1 — Electric Charges and Fields

60. Sketch the electric field lines for (a) a positive point charge and (b) an electric dipole. (Pg 8 and 9)
61. Define electric flux through an area element. Mention its SI unit. (Pg 10)
62. Define electric field at a point. What is meant by 'source charge'?
63. Define electric dipole moment and write its SI unit. (Pg 10)
64. When is the torque on an electric dipole in a uniform electric field (a) maximum and (b) minimum? (Pg 13)

Chapter 2 — Electrostatic Potential and Capacitance

65. What is meant by equipotential surface? Draw equipotential surfaces for an electric dipole. (Pg 35)
66. Mention any two properties of equipotential surface. (Pg 35)
67. What are polar molecules? Give an example. (Pg 42)
68. What are non-polar molecules? Give an example. (Pg 42)
69. Write any two differences between polar and non-polar molecules. (Pg 43)
70. The electrostatic potential energy of a system of two like charges is positive. Explain.
71. What is electrostatic shielding? Mention one use of it.

Chapter 3 — Current Electricity

72. State and explain Ohm's law. (Pg 65)
73. How does resistance of a conductor depend on length and area of cross-section of a conductor? (Pg 65)
74. Mention any three factors on which resistance of a conductor depends. (Pg 66)
75. Write the expression for drift velocity in terms of current and explain the terms. (Pg 68)
76. Define free electron mobility and mention its SI unit. (Pg 71)
77. What are ohmic and non-ohmic devices? (Pg 72)
78. Write any two limitations of Ohm's law. (Pg 73)
79. State Kirchhoff's rules of an electric network. (Pg 78)
80. Name the mobile charge carriers in metals. Write the expression for mobility in terms of average collision time (relaxation time) for charge carriers.
81. How does the resistivity of a semiconductor vary if its temperature is increased? Also show the variation graphically.

Chapter 4 — Moving Charges and Magnetism

82. A current loop of magnetic dipole moment 10 Am^2 is freely suspended in a uniform magnetic field of $5 \times 10^{-3} \text{ T}$ and its plane is present in the direction of magnetic field. Calculate the torque that acts on the current loop.
83. Give the vector form of Biot-Savart's law and explain the terms. (Pg 94)
84. What is the nature of the force between two parallel conductors carrying currents in the (a) same direction and (b) opposite direction? (Pg 97)
85. When does a charged particle moving in a uniform magnetic field experience (a) maximum force and (b) minimum force? (Pg 90)

86. When does a straight current carrying conductor placed in a uniform magnetic field experience (a) maximum force and (b) minimum force? (Pg 91)
87. When is the torque acting on a rectangular current loop placed in a uniform magnetic field (a) maximum and (b) minimum? (Pg 101)
88. Write any two reasons to show that a galvanometer as such cannot be used as an ammeter. (Pg 102)

Chapter 6 — Electromagnetic Induction

89. State and explain Faraday's law of electromagnetic induction. (Pg 116)
90. Write the expression for the motional emf induced in a straight conductor moving perpendicular to a uniform magnetic field. Explain the terms.
91. When is magnetic flux through a surface (a) maximum and (b) minimum? (Pg 117)
92. Mention the two factors on which self-inductance of a solenoid depends. (Pg 124)
93. What is an AC generator? What is its working principle? (Pg 126 and Pg 127)
94. Calculate the magnetic energy stored in a coil of self-inductance 0.05 H carrying a current of 0.2 A.

Chapter 8 — Electromagnetic Waves

95. What is displacement current? Write the mathematical form of Ampere-Maxwell law. (Pg 153)
96. Write the expression of displacement current and explain the terms. (Pg 153)
97. List any two uses of infrared rays. (Pg 155)
98. List any two uses of microwaves. (Pg 155)
99. List any two uses of UV rays. (Pg 156)
100. List any two uses of X-rays. (Pg 156)
101. Name the electromagnetic waves used in: (i) The radar systems of aircraft. (ii) The laser assisted eye surgery (LASIK).

Chapter 12 — Atoms

102. Define impact parameter. What is the angle of scattering for minimum impact parameter?
103. The radius of the innermost electron orbit of a hydrogen atom is 5.3×10^{-11} m. Determine the radius of the $n = 2$ orbit.
104. Give any two limitations of Bohr's atomic model. (Pg 78)
105. State the postulates of Bohr's theory of hydrogen atom. (Pg 75)

Chapter 14 — Semiconductor Electronics

106. Give any two differences between intrinsic and extrinsic semiconductors. (Pg 101)
107. Mention any two differences between half-wave and full-wave rectifiers.
108. What is forward bias and reverse bias? (Pg 106 and 107)
109. List out two differences between p-type and n-type semiconductors. (Pg 101)

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